Exponential Positive-Evidence Requirements for Two-Tier Local Languages

Small Syntactic Monoids, Hypergraph Transversals, and Exact Sampling Bounds

October 7, 2026

Abstract

We construct strictly 2-local languages $L_{r,d}$, over $dr + 2$ symbols, whose optimal positive characteristic-sample cardinality is d^2 in the single-tier hypothesis class but exactly d^r when two nested tiers are permitted, for $r \geq 2$ and $d \geq 2$. Every two-tier characteristic sample contains all d^r designated words and has total length at least $(r + 2)d^r$. The targets have $r + 4$ minimal deterministic states and $\mathcal{J}$-trivial syntactic monoids of exactly $(r + 2)(r + 3)/2 + 2$ elements, satisfying $x^2 = x^3$. Their canonical multitier grammars have polynomial size. Thus uniform polynomial characteristic-data bounds fail even for two nested tiers and small algebraic representations.

A complementary construction realizes an exact weighted hypergraph-transversal formula for characteristic length. With r forbidden tier-bigrams and at most t blocker symbols per constraint, its worst-case value is $(r + 2)t^r$. For the layered target and its single-word deletions, independent uniform positive sampling has minimax exact-identification complexity

$$\Theta(d^r\,(r \log d + \log(1/\delta)))\,,$$

although this finite family has VC and Littlestone dimension one. The characteristic obstruction transfers to partial identity transductions. These results distinguish restrictions on targets from restrictions on competing hypotheses and give explicit counterexamples to uniform polynomial-data assertions for unrestricted multitier string learning.

1 Introduction and literature comparison

The qualitative learnability of bounded subregular classes is not an open starting point. The quantitative questions depend on what is counted, which hypotheses are permitted, and how positive evidence arrives. In particular,

(i) a small characteristic sample need not appear quickly in random data;

(ii) a small number of examples can conceal exponentially long words;

(iii) a small target automaton need not imply a small positive certificate;

(iv) restricting a target's representation does not restrict its competitors;

(v) exact identification, ordinary teaching, and approximate prediction have different information requirements.

The main question addressed here is the following.

Problem 1.1 (Uniform positive characteristic data)**.** With the alphabet supplied but allowed to vary, is there a learner for two-tier strictly 2-local languages whose positive characteristic samples are polynomially bounded in alphabet size, target grammar size, minimal DFA size, and syntactic-monoid size? Does restricting the tiers to be nested, or the syntactic monoid to be $\mathcal{J}$-trivial, restore such a bound?

The answer is negative. The target itself may be strictly local. Its characteristic complexity changes exponentially when the hypothesis class admits certain two-tier refinements.

1.1 Established learning results

Angluin's characterization of positive-data identification provides the telltale framework, while de la Higuera develops the quantitative characteristic-sample criterion [1, 2]. The strong batch characteristic samples used below require correctness on every target-consistent extension of the sample. They are not teacher-encoded sequences.

Heinz's string-extension learning and the lattice framework of Heinz, Kasprzik, and Kötzing already supply the finite-feature learners underlying fixed-locality SL and SP inference [3, 4]. Our feature-cover lemma is a quantitative specialization of this established mechanism, not a new qualitative learning algorithm.

Jardine and Heinz prove efficient learning of TSL_2; Jardine and McMullin extend the result to supplied locality k [5, 6]. The latter's constant-data formulation fixes both alphabet and locality. It does not provide a constant uniform in those parameters. The exact single-tier calculation below concerns a special finite target family and does not replace these more general algorithms.

1.2 The closest multitier assertion

The closest direct claim is Swanson's MTSL-BUFIA theorem [7]. Its Lemma 2 asserts polynomial representative data for unrestricted MTSL and gives an $O(|\Sigma|^2 + |G|)$ estimate, counting symbols in tiers and factors. Theorem 1 then concludes identification with polynomial data. Our counterexamples concern precisely the uniform, varying-alphabet interpretation of that assertion. They do not dispute qualitative convergence.

The later, tree-oriented treatment [8] repeats an analogous polynomial-data assertion in Lemma 5. The comparison proved here is confined to the string setting of [7]; no theorem about unrestricted tree presentations is inferred without an additional encoding argument.

1.3 Algebra, transductions, and noise

Algebraic characterizations of SP, SL, and related strict testability notions are established [9, 10]. In particular, plain $\mathcal{J}$-triviality must not be equated with strict piecewise acceptance. Our contribution is an explicit learning obstruction with a fully computed small syntactic monoid, not a replacement structural characterization.

Polynomial-time-and-data learning is also established for input strictly local functions, output strictly local functions, and restricted multiple independent tier-based processes [12, 13, 14]. Our transduction lower bound concerns unknown multitier domains. A separate known-controller proposition isolates the simpler transition-coverage problem.

Wu and Heinz study noisy string-extension learning under frequency separation assumptions [15]. Our finite-sample noise bounds use an explicit probabilistic version of such separation; arbitrary contamination is treated separately.

Hypergraph transversal duality is classical [11]. No novelty is claimed for that duality, finite-feature coverage, or coupon collection in isolation. The results advanced here are their exact realization in the stated language classes, the two-nested-tier separation, the small-monoid calculation, and the matching-order statistical obstruction. The literature comparison is scoped to these statements; it is not an assertion of exhaustive historical priority for every auxiliary formulation.

2 Language classes and learning models

2.1 Boundary conventions

Let Σ be a finite alphabet, $s = |\Sigma| \geq 1$. For a word w, let $\mathrm{Fac}_k(w)$ denote its contiguous factors of length k. For a set W, let $\mathrm{Fac}(W)$ contain all factors of members of W, including ε.

For $T \subseteq \Sigma$, let

$$\pi_T : \Sigma^* \longrightarrow T^*$$

erase letters outside T.

Definition 2.1 (Boundary-free classes)**.** For $F \subseteq T^k$, define

$$L(T, F) = \{w : \mathrm{Fac}_k(\pi_T(w)) \cap F = \varnothing\}.$$

The class with $T = \Sigma$ is $\mathsf{SL}_k^\circ(\Sigma)$; the class with one arbitrary tier is $\mathsf{TSL}_k^\circ(\Sigma)$.

Let $\mathcal{C}_{\leq t}(\Sigma)$ consist of intersections of at most t members of $\mathsf{TSL}_2^\circ(\Sigma)$, and let $\mathcal{C}_{\mathrm{nest}}$ restrict the two-tier case to nested tiers. The unrestricted finite-intersection closure is $\mathsf{MTSL}_2^\circ(\Sigma)$.

The superscript $^\circ$ explicitly excludes boundary conditions. These are subclasses of the usual boundary-sensitive classes. Our lower bounds persist in the larger classes because all targets and distinguishing competitors remain available. Exact upper bounds are not transferred silently across conventions.

An atomic bigram constraint is (T, xy), where $x, y \in T$. Its language forbids xy on tier T. For an explicit collection G of atoms, define

$$g(G) = \sum_{(T,xy)\in G} (|T| + 2).$$

A binary encoding adds at most an $O(\log s)$ factor for symbol names.

Every boundary-free multitier language is *factorial*: if uwv is accepted, then w is accepted. This follows because $\pi_T(w)$ is a factor of $\pi_T(uwv)$.

For later comparison, write $u \preceq w$ when u is a subsequence of w, and define

$$\mathrm{Sub}_{\leq k}(w) = \{u \in \Sigma^{\leq k} : u \preceq w\}.$$

An SP_k language avoids a specified set of such subsequences. Allowing the empty forbidden subsequence includes the empty language.

2.2 Characteristic samples and telltales

A positive sample is a finite set $S \subseteq L$. We distinguish

$$|S|, \qquad \ell(S) = \sum_{w\in S} |w|, \qquad \ell_+(S) = \sum_{w\in S} (1 + |w|).$$

A positive text exhaustively enumerates the target, with repetitions permitted. For an empty target, one may use a pause symbol that contributes no word to the sample.

Definition 2.2. A batch learner A identifies a class from positive texts if its hypotheses eventually stabilize extensionally to the target on every positive text.

A finite $S \subseteq L$ is *characteristic* for A, L if

$$S \subseteq D \subseteq L, \quad D \text{ finite} \quad \Longrightarrow \quad A(D) = L.$$

Let $\chi_{\mathcal{C}}(L)$, respectively $\lambda_{\mathcal{C}}(L)$, be the minimum possible characteristic cardinality, respectively total symbol length, among learners identifying $\mathcal{C}$ and having characteristic samples throughout $\mathcal{C}$.

There is no computational restriction in these minima.

Definition 2.3. A positive *telltale* for L relative to $\mathcal{C}$ is a finite $S \subseteq L$ such that

$$S \subseteq H \subseteq L, \quad H \in \mathcal{C} \quad \Longrightarrow \quad H = L.$$

Lemma 2.4 (Necessary positive distinctions)**.** *Every characteristic sample for a successful positive-text learner is a telltale. In particular, if*

$$L \setminus \{w\} \in \mathcal{C} \qquad (w \in W \subseteq L),$$

every characteristic sample for L contains W. Every ordinary labeled teaching set for L also contains each $w \in W$, labeled positive.

Proof. If a characteristic sample S is contained in a proper sublanguage $H \in \mathcal{C}$, take a text for H beginning with S. Every subsequent sample extends S and is contained in L. The learner must output L forever, contradicting identification of H.

For labeled teaching, L and $L \setminus \{w\}$ disagree only at w. No other labeled observation distinguishes them. □

Ordinary positive teaching requires uniqueness among *all* consistent hypotheses, including superlanguages. Since Σ^* belongs to our full classes, no proper target has an ordinary positive-only teaching set. This does not contradict finite characteristic samples.

Lemma 2.5 (Finite-class learners)**.** *Every finite class of regular languages has a characteristic-complete positive learner. If the class is factorial and $L = \mathrm{Fac}(W)$ for finite $W \subseteq L$, then W is characteristic for such a learner.*

Proof. Order the distinct languages by a linear extension of proper inclusion, smaller languages first, and output the first consistent one. Every language preceding a target L omits a word of L. Finitely many such witnesses eliminate all earlier hypotheses and form a characteristic sample.

If a factorial hypothesis contains W, it contains $\mathrm{Fac}(W) = L$. Thus after W is observed, no earlier hypothesis can be consistent. □

For fixed alphabet and locality, all classes used below are finite. The issue is uniform quantitative dependence on their parameters.

3 The finite-feature principle

Let U be a finite feature set and $\phi : \Sigma^* \to 2^U$. Put

$$L_B = \{w : \phi(w) \subseteq B\}, \qquad \mathcal{C}_\phi = \{L_B : B \subseteq U\},$$

and write

$$\phi(S) = \bigcup_{w \in S} \phi(w).$$

Proposition 3.1 (Standard coverage characterization)**.** *For $L \in \mathcal{C}_\phi$,*

$$L = L_{\phi(L)}.$$

A positive sample $S \subseteq L$ is a telltale if and only if $\phi(S) = \phi(L)$. Consequently,

$$\chi_{\mathcal{C}_\phi}(L) = \min\{|S| : S \subseteq L,\ \phi(S) = \phi(L)\},$$

with the analogous equality for λ. The learner $S \mapsto L_{\phi(S)}$ attains these minima.

Proof. If $L = L_B$, then $\phi(L) \subseteq B$, and both inclusions in $L = L_{\phi(L)}$ follow directly.

The language $L_{\phi(S)}$ is the least class member containing S, and is contained in L. Thus a telltale forces $L_{\phi(S)} = L$, which forces equality of feature unions. Conversely, equality of feature unions forces every consistent sublanguage to equal L.

Once a sample covers $\phi(L)$, every positive extension does too. Choosing one positive witness per live feature proves existence. Lemma 2.4 gives the learner-independent lower bound. □

For MTSL_2°, take

$$U_\Sigma = \{(T, xy) : T \subseteq \Sigma,\ x, y \in T\},$$
$$\phi(w) = \{(T, xy) : xy \in \mathrm{Fac}_2(\pi_T(w))\}.$$

Then

$$|U_\Sigma| = \sum_{T \subseteq \Sigma} |T|^2 = s(s+1)2^{s-2}. \tag{1}$$

Repeated-letter pairs contribute $s2^{s-1}$; distinct ordered pairs contribute $s(s-1)2^{s-2}$. This proves a finite, generally exponential, characteristic upper bound. It does not prove a polynomial bound in a compact grammar.

4 An exact separation between one tier and two nested tiers

4.1 Layered targets

Fix $r \geq 1$ and $d \geq 2$. Let

$$B_0 = \{a\}, \qquad B_{r+1} = \{b\},$$

and let $B_1, \ldots, B_r$ be disjoint d-element alphabets, also disjoint from $\{a, b\}$. Set

$$\Sigma_{r,d} = \bigcup_{i=0}^{r+1} B_i, \qquad s = dr + 2,$$

$$W_{r,d} = B_0 B_1 \cdots B_r B_{r+1}, \qquad L_{r,d} = \mathrm{Fac}(W_{r,d}).$$

Write $K = d^r = |W_{r,d}|$.

A nonempty accepted word uses a consecutive interval of layers, one letter per layer. Hence

$$L_{r,d} = \{w : \mathrm{Fac}_2(w) \subseteq E_{r,d}\}, \qquad E_{r,d} = \bigcup_{i=0}^{r} B_i B_{i+1}.$$

In particular,

$$L_{r,d} \in \mathsf{SL}_2^\circ, \qquad |E_{r,d}| = 2d + (r-1)d^2, \tag{2}$$

and a full-tier grammar uses

$$f_{r,d} = s^2 - 2d - (r-1)d^2$$

forbidden bigrams.

Lemma 4.1 (One additional tier deletes one word)**.** *For $w = ax_1 \cdots x_r b \in W_{r,d}$, define*

$$T_w = \{a, b\} \cup \bigcup_{i=1}^{r} (B_i \setminus \{x_i\}).$$

Then

$$L_{r,d} \cap L(T_w, \{ab\}) = L_{r,d} \setminus \{w\}.$$

Thus every single-word deletion belongs to $\mathcal{C}_{\text{nest}}$.

Proof. A word of $L_{r,d}$ containing both a and b must be a full path $ay_1 \cdots y_r b$. It has adjacent ab after projection to T_w precisely when all interior letters are erased, namely when $y_i = x_i$ for every i. Words missing an endpoint cannot violate the added constraint. The two tiers are $T_w \subseteq \Sigma_{r,d}$. □

Theorem 4.2 (Exact one-tier versus two-tier characteristic costs)**.** *For $r \geq 2, d \geq 2$,*

$$\chi_{\mathsf{SL}_2^\circ}(L_{r,d}) = \chi_{\mathsf{TSL}_2^\circ}(L_{r,d}) = d^2,$$

whereas

$$\chi_{\mathcal{C}_{\text{nest}}}(L_{r,d}) = \chi_{\mathcal{C}_{\leq 2}}(L_{r,d}) = \chi_{\mathsf{MTSL}_2^\circ}(L_{r,d}) = d^r.$$

For each of the latter three classes,

$$\lambda(L_{r,d}) = (r+2)d^r,$$

and the minimum delimiter-counted size is $(r+3)d^r$. The sample $W_{r,d}$ attains all these latter minima.

For $r = 1$, both one-tier and multitier cardinalities are d.

Proof. In the local class, every allowed edge between B_1 and B_2 must occur in a characteristic sample: otherwise forbidding that edge gives a proper consistent local sublanguage. There are d^2 such edges, and one positive word contains at most one. This proves the lower bound when $r \geq 2$.

Index $B_i = \{x_{i,1}, \ldots, x_{i,d}\}$. For every $(u, v) \in [d]^2$, take the full path using index u on odd internal layers and index v on even internal layers. These d^2 words cover every allowed bigram, including the initial and final ones. Proposition 3.1 proves sufficiency for SL_2°.

For the single-tier upper bound, any language described on a proper tier is infinite: if c is omitted from the tier, every c^j projects to ε and is accepted. Thus every finite TSL_2° language is a full-tier local language.

Choose an inclusion-respecting order of this finite hypothesis class with all finite languages before all infinite languages. A finite hypothesis containing the d^2 selected words permits every edge in $E_{r,d}$, and therefore contains $L_{r,d}$. The first-consistent learner consequently has those words as a characteristic sample for $L_{r,d}$. The local lower bound remains applicable.

For the multitier classes, Lemmas 4.1 and 2.4 force every member of $W_{r,d}$ into every characteristic sample. Conversely, all hypotheses are factorial, and $L_{r,d} = \text{Fac}(W_{r,d})$. Lemma 2.5 makes $W_{r,d}$ characteristic, including in the nested two-tier class. Each required word has length $r + 2$, proving the exact length costs.

When $r = 1$, the d initial edges require d examples, and the d full paths suffice in every class under consideration. □

For $d = 2$, the one-tier cardinality is four and the two-tier cardinality is 2^r. The separation is strict for $r \geq 3$.

4.2 The canonical grammar is polynomial

For each endpoint pair xy, retain all inclusion-minimal tiers containing its letters on which xy never occurs in the target. Call the resulting collection $G_{\min}(L)$. Minimality is taken separately for each endpoint pair; constraints with different endpoints may be logically redundant.

Proposition 4.3 (Canonical grammar calculation)**.** *Let $x \in B_i$, $y \in B_j$. The minimally forbidding tiers for xy in $L_{r,d}$ are:*

$$\begin{cases} \{\mathrm{alph}(xy)\}, & j \le i, \\ \varnothing, & j = i+1, \\ \big\{\{x,y\} \cup B_h : i < h < j\big\}, & j \ge i+2. \end{cases}$$

These constraints define $L_{r,d}$, and

$$g(G_{\min}(L_{r,d})) \le s^2\big(4 + (d+4)r\big).$$

For $r \ge 2$, the canonical grammar of $L_{r,d} \setminus \{w\}$ is obtained by adding exactly (T_w, ab), increasing its size by $r(d-1)+4$.

Proof. If $j \le i$, no accepted word places x before y, so the endpoint tier already forbids the pair. If $j = i+1$, the positive word xy rules out every forbidding tier.

Suppose $j \ge i+2$. Every accepted occurrence from x to y contains exactly one letter from each intermediate layer. A tier separates the endpoints in every such occurrence if it contains an entire intermediate layer. Conversely, if it contains no entire intermediate layer, choose an omitted symbol from each one; the resulting positive word projects to xy. This proves the list and its minimality.

A word outside $L_{r,d}$ has an adjacent pair whose layer indices are not consecutive increasing indices. Such a pair violates one of the listed constraints, proving that the canonical grammar defines the target.

There are at most s^2 constraints of the first kind, of cost at most four, and at most rs^2 of the third kind, of cost at most $d+4$. This gives the bound.

After deleting a full path w, every endpoint configuration other than ab occurring in w remains witnessed by a proper factor of w. Only ab can acquire new constraints.

For ab, a tier containing no entire internal layer fails precisely on the full paths obtained by independently choosing a symbol omitted from each layer. It can remove only w exactly when the unique omitted symbol in each layer is the corresponding symbol of w. This gives T_w. For $r \ge 2$, the original tiers remain minimal, because changing a choice in another layer supplies a surviving witness. □

Thus, for fixed d, the canonical grammar has size $O(r^3)$. For $d = 2$, every canonical target tier has at most four symbols.

5 Exact automata and syntactic monoids

Recall

$$u \equiv_L v \quad \Longleftrightarrow \quad \forall p, q \in \Sigma^* \; [puq \in L \Longleftrightarrow pvq \in L].$$

The quotient is $\mathrm{Syn}(L)$. For a monoid M,

$$x \mathrel{\mathcal{J}} y \quad \Longleftrightarrow \quad MxM = MyM.$$

A monoid is $\mathcal{J}$-trivial if this implies $x = y$.

Theorem 5.1 (Exact algebraic size)**.** *The minimal complete DFA of $L_{r,d}$ has*

$$n = r + 4$$

states. Put $m = r + 2$. Its syntactic monoid has

$$|\operatorname{Syn}(L_{r,d})| = \frac{m(m+1)}{2} + 2$$

elements: an identity, a zero, and elements

$$e_{ij}, \qquad 0 \le i \le j \le m-1,$$

with multiplication

$$e_{ij}e_{h\ell} = \begin{cases} e_{i\ell}, & h = j+1, \\ 0, & h \neq j+1. \end{cases}$$

It is $\mathcal{J}$-trivial. Every nonidentity element has square zero, so the monoid satisfies $x^2 = x^3$.

Proof. Use an initial state q_*, layer states $q_0, \dots, q_{r+1}$, and a sink $\perp$. All states except $\perp$ accept. From q_*, a letter in B_i goes to q_i. From q_i, only a letter in B_{i+1} avoids the sink, and it goes to q_{i+1}. This recognizes the consecutive-layer words.

All states are reachable. The sink is distinguished by the empty continuation. The initial state accepts every one-letter continuation, whereas a layer state accepts only letters of the next layer, or none at the final layer. Distinct layer states therefore have distinct right languages. The DFA is minimal.

A nonempty valid word spans an interval $[i, j]$ of layers. It induces a transformation sending q_*, and also q_{i-1} when $i > 0$, to q_j, and sending every other nonsink state to $\perp$. These transformations are distinct. An invalid word sends every state to $\perp$. The empty word induces the identity. Concatenation gives exactly the displayed multiplication.

For a minimal DFA, equality of word transformations is equivalent to syntactic equivalence: equality of transformations gives equality under every left and right context, and distinct reachable-state images can be separated by a right continuation. Thus the counted transition monoid is the syntactic monoid.

The two-sided principal ideal of e_{ij} is

$$\{0\} \cup \{e_{h\ell} : h \le i,\ \ell \ge j\}.$$

Different intervals have different ideals, while identity and zero also have distinct ideals. This proves $\mathcal{J}$-triviality. Finally $i \le j$ excludes $i = j + 1$, so $e_{ij}^2 = 0$. □

The competitors are also algebraically small.

Proposition 5.2 (Complexity of the single-deletion competitors)**.** *Every $L_{r,d} \setminus \{w\}$, $w \in W_{r,d}$, has a complete DFA with at most $(r+4)^2$ states and a syntactic monoid of size at most*

$$\left(\frac{(r+2)(r+3)}{2} + 2\right)^2.$$

Its syntactic monoid is $\mathcal{J}$-trivial and satisfies $x^2 = x^3$.

Proof. Intersect the target DFA with the complement of the singleton-word DFA for w. A word of length $r+2$ has a complete singleton DFA with $r+4$ states.

Every letter of w is distinct. A nonempty word induces a nonzero transformation of its singleton DFA only if it is a factor of w. Each factor has a unique occurrence and induces a distinct one-source transformation. Including identity and zero, this monoid has $(r+2)(r+3)/2+2$ elements. Complementing final states does not change it. The product with the target monoid recognizes the deletion language, so its syntactic monoid is a quotient of a submonoid of that product and has the stated size bound.

Every nonempty squared word is outside $L_{r,d}$, in every context, because its repeated layers cannot form a consecutive increasing interval. Its syntactic image in the deletion language is therefore zero. No nonempty word represents the identity of this nonempty finite language, since that would force infinitely many powers into the language. Thus $x^2 = x^3$.

For completeness, every finite language has a $\mathcal{J}$-trivial syntactic monoid. If nonzero elements satisfy $x = ayb$ and $y = cxd$, then $x = acxdb$. If any multiplier is nonidentity, choose word representatives and iterate this context to obtain arbitrarily long words in the same nonzero syntactic class. A fixed accepting context would then produce arbitrarily long accepted words, contradicting finiteness. All multipliers must therefore be identities, giving $x = y$. □

Corollary 5.3 (Negative answer to Problem 1.1)**.** *No polynomial in alphabet size, target grammar size, minimal DFA size, and syntactic-monoid size uniformly bounds positive characteristic cardinality or length for two nested tiers. The obstruction persists when both the target and its distinguishing competitors have polynomial-size automata and monoids, are $\mathcal{J}$-trivial, and satisfy $x^2 = x^3$.*

Proof. Set $d = 2$ and let r grow. All displayed structural sizes are polynomial in r, whereas Theorem 4.2 forces 2^r positive words. □

Remark 5.4 (Acceptance structure and hypothesis dependence). This does not identify SP with $\mathcal{J}$-triviality. For example, $L_{r,d}$ is not subsequence downward closed: $ax_1x_2 \in L_{r,d}$, but $ax_2 \notin L_{r,d}$.

More importantly, the *same* target, with the same accepting automaton and syntactic monoid, has characteristic cardinality d^2 in one hypothesis class and d^r in another. A target invariant cannot determine a class-relative learning complexity without specifying the competitors.

6 The combinatorial obstruction: minimal interveners

6.1 An exact criterion for unrestricted multitier learning

For $x, y \in \Sigma$, an xy-occurrence is a factor xvy whose interior contains neither endpoint symbol. When $x = y$, the endpoints are consecutive occurrences of that symbol. Define

$$\mathcal{I}_{xy}(L) = \{\mathrm{alph}(v) : xvy \text{ is such an occurrence in a word of } L\},$$

$$\mathcal{A}_{xy}(L) = \mathrm{Min}_{\subseteq}\, \mathcal{I}_{xy}(L), \qquad h(L) = \sum_{x,y\in\Sigma} |\mathcal{A}_{xy}(L)|.$$

Proposition 6.1 (Minimal-intervener coverage)**.** *For $L \in \mathsf{MTSL}_2^\circ$, a positive sample covers all target tier features if and only if, for every x, y and every $I \in \mathcal{A}_{xy}(L)$, it contains an xy-occurrence with intervener set exactly I. Consequently,*

$$\chi_{\mathsf{MTSL}_2^\circ}(L) \le h(L).$$

If L has an n-state DFA, the witnesses can be chosen so that

$$\ell(S) \le (3n-1)h(L).$$

Proof. For $I \in \mathcal{A}_{xy}(L)$, put $T = \Sigma \setminus I$. The target realizes (T, xy). A sample witness for that feature has an intervener set $J \subseteq I$. Since it is positive, minimality forces $J = I$.

Conversely, a realized feature (T, xy) has some target intervener set J disjoint from T. Choose a minimal intervener set $I \subseteq J$. A sample occurrence with exact support I witnesses the same feature.

The occurrence property $xy \in \mathrm{Fac}_2(\pi_T(w))$ has a three-state DFA: no pending x, pending x, and occurrence found. Non-tier symbols leave the state unchanged. Intersecting it with the target DFA gives at most $3n$ states. A shortest accepted witness has length at most $3n - 1$. Minimality of I again ensures an exact-support occurrence. □

For $s \geq 2$, these minimal sets form antichains on at most $s - 2$ symbols for distinct endpoints and $s - 1$ symbols for equal endpoints. Hence

$$h(L) \leq s(s-1)\binom{s-2}{\lfloor (s-2)/2 \rfloor} + s\binom{s-1}{\lfloor (s-1)/2 \rfloor}. \tag{3}$$

Indeed, for an antichain $\mathcal{A} \subseteq 2^{[m]}$, count permutations whose first $|A|$ elements form A. No permutation is counted twice, yielding

$$\sum_{A \in \mathcal{A}} \binom{m}{|A|}^{-1} \leq 1,$$

and therefore $|\mathcal{A}| \leq \binom{m}{\lfloor m/2 \rfloor}$.

In the layered construction,

$$\mathcal{A}_{ab}(L_{r,d}) = \big\{\{x_1, \ldots, x_r\} : x_i \in B_i\big\}.$$

There are d^r minimal interveners, although only r minimal forbidding tiers for ab.

6.2 An exact weighted-transversal realization

A second construction isolates the number of forbidden constraints. Let B be a finite alphabet disjoint from a, b. For $E \subseteq B$, put

$$P_E = \{w : ab \notin \mathrm{Fac}_2(\pi_{\{a,b\} \cup E}(w))\}.$$

For a hypergraph $H \subseteq 2^B$, define

$$Q_H = \bigcap_{E \in H} P_E, \qquad \mathcal{Q}_B = \{Q_H : H \subseteq 2^B\}.$$

A transversal meets every edge of H; let $\mathrm{Tr}(H)$ denote the inclusion-minimal transversals. Use the conventions

$$\mathrm{Tr}(\varnothing) = \{\varnothing\}, \qquad \varnothing \in H \Longrightarrow \mathrm{Tr}(H) = \varnothing.$$

An ab-interval is an occurrence avb with $v \in B^*$. A word belongs to Q_H exactly when every such interval has interior alphabet meeting every edge of H. Distinct ab-intervals have disjoint positions: their interiors contain neither endpoint, and a right endpoint b cannot also be a left endpoint a.

Theorem 6.2 (Exact weighted-transversal formula)**.** *For every $H \subseteq 2^B$,*

$$\lambda_{\mathcal{Q}_B}(Q_H) = \sum_{W \in \mathrm{Tr}(H)} (|W| + 2).$$

A sample is a telltale exactly when it contains an ab-interval with interior alphabet W for every $W \in \mathrm{Tr}(H)$. If $\mathrm{Tr}(H) \neq \varnothing$, the minimum length is attained by one word. If $\mathrm{Tr}(H) = \varnothing$, the empty sample suffices.

If H has $r \geq 1$ nonempty edges of size at most $t \geq 1$, then

$$\lambda_{\mathcal{Q}_B}(Q_H) \leq (r+2)t^r.$$

Equality holds for r disjoint t-element edges. The comparison class in all these statements permits arbitrary additional blocker sets.

Proof. Suppose a minimal transversal W is absent from the sample's interval supports. Add the constraint $P_{B\setminus W}$. It excludes a word $av_W b$, where v_W lists W, so gives a proper sublanguage. A positive sample word violating this additional constraint would have an interval support $U \subseteq W$. Since U is a transversal, minimality forces $U = W$, a contradiction. Thus every minimal transversal must be represented.

Conversely, suppose all are represented. Any constraint P_E consistent with the sample has E meeting every minimal transversal. It then meets every transversal, because every transversal contains a minimal one. The constraint is therefore valid throughout Q_H. The least consistent conjunction equals Q_H, proving sufficiency.

A required interval with support W costs at least $|W| + 2$ symbols. The disjointness of intervals proves the sum lower bound. Concatenate one block $av_W b$ per minimal transversal. At block junctions the endpoint pattern is ba, so no extra ab-interval is created. This single positive word attains the sum. If an empty edge is present, $Q_H = P_\varnothing$ is the least member of $\mathcal{Q}_B$, and needs no positive telltale.

For the final bound, every x in a minimal transversal W has a private edge E_x satisfying $E_x \cap W = \{x\}$. Different x's require different private edges, so $|W| \leq r$.

Fix an order on B. Map W to the tuple whose coordinate for edge E is the least element of $W \cap E$. Every $x \in W$ appears at its private edge, so the tuple determines W. There are at most t^r tuples. For disjoint t-element edges, the minimal transversals choose one element from each edge, giving t^r sets of size r. □

Corollary 6.3 (Linear grammars with exponential symbol requirements)**.** *Take r disjoint two-element edges. The target Q_H has alphabet size $2r + 2$ and canonical grammar size $6r$, using exactly r forbidden bigrams on four-symbol tiers. Every characteristic sample for a learner identifying unrestricted MTSL_2° has length at least*

$$(r+2)2^r.$$

The unrestricted feature learner has a characteristic sample of length at most

$$(r+2)2^r + 2\big((2r+2)^2 - 1\big).$$

Proof. Every two-letter word except ab is positive, so no other endpoint pair has a valid prohibition. For ab, a blocker set E meets every transversal precisely when it contains an edge of H: otherwise choose one symbol from each edge outside E, obtaining a transversal disjoint from E. Thus the displayed disjoint edges are exactly the canonical blockers.

The additional constraints used in Theorem 6.2 are available in the unrestricted class, giving its lower bound. For the upper bound, use the single concatenated witness and every positive two-letter word. The latter refute every constraint on a pair other than ab, and the former refutes every invalid ab-constraint. □

This construction is complementary to the layered one. It gives a linear canonical grammar and an exact exponential *symbol* requirement, but one long word can encode all witnesses. The layered construction prevents such packing.

The sparse family is also aperiodic with a fixed exponent. A single P_E is recognized by a three-state chain $0 < 1 < \bot$, where a sets a pending endpoint, b moves a pending endpoint to $\bot$, and a blocker resets it. All letter transformations preserve this order. Every order-preserving map of a three-element chain satisfies $x^2 = x^3$, because each orbit stabilizes after at most two strict moves. Products and quotients preserve the identity, so $\mathrm{Syn}(Q_H)$ satisfies it for every H.

For disjoint nonempty edges, the minimal complete DFA of Q_H has exactly $2^r + 1$ states. Use subsets of $[r]$, all accepting, and a rejecting sink. Start at $[r]$; a resets to $\varnothing$, a blocker from group i adds i, and b is permitted only at $[r]$. All subsets are reachable. For distinct subsets S, T, choose $i \in S \setminus T$, after interchanging them if necessary, then append blockers from groups outside S, followed by b. This accepts from S and rejects from T. Thus this second family does not independently give an exponential lower bound in DFA size.

7 The precise comparison with previous polynomial-data bounds

The uniform representative-data estimate in [7, Lemma 2] cannot hold for unrestricted multitier string languages with the stated symbol-count representation. There are two separate counterexamples:

1. $L_{r,2}$ has a polynomial-size canonical grammar, but every characteristic sample contains 2^r distinct words.

2. The disjoint-blocker language in Corollary 6.3 has canonical grammar size $6r$, but needs at least $(r + 2)2^r$ symbols.

The proof issue is combinatorial, not computational. The cited argument treats small witnesses for individual tier members as sufficient to exclude all spurious forbidding tiers. Its representative-sample definition also requires exclusion of those spurious tiers; that additional condition is not implied by the individual witnesses.

For a concrete example, let

$$H = \{\{x_1, y_1\}, \ldots, \{x_r, y_r\}\}.$$

Consider the $r + 1$ supports

$$W_0 = \{x_1, \ldots, x_r\}, \qquad W_i = (W_0 \setminus \{x_i\}) \cup \{y_i\}.$$

They individually witness each blocker without its partner. For $r \geq 2$, however, they omit the minimal transversal choosing y_1, y_2 and x_j for $j > 2$. The constraint indexed by its complement remains consistent. Adding every positive surface bigram does not remove this ambiguity.

No larger data-independent constant repairs an exponential versus polynomial discrepancy when the alphabet varies. If the alphabet is fixed externally, the entire feature universe is fixed; a bound with arbitrary alphabet-dependent constants is a different statement. Neither qualitative identification nor the specialized single-tier results are contradicted.

8 Teaching, dimensions, and active learning

Let

$$L = L_{r,d}, \qquad W = W_{r,d}, \qquad K = d^r,$$

and define the single-deletion family

$$\mathcal{H} = \{L\} \cup \{L \setminus \{w\} : w \in W\}.$$

All members have polynomial-size descriptions for fixed d.

The VC dimension is the largest cardinality of a shattered set. The Littlestone dimension is the largest depth of a complete binary instance tree for which every root-to-leaf labeling is realized.

Proposition 8.1 (Small dimensions, large exact teaching requirements)**.** *For $K \geq 2$,*

$$\mathrm{VCdim}(\mathcal{H}) = \mathrm{Ldim}(\mathcal{H}) = 1, \qquad \mathrm{TD}_{\mathcal{H}}(L) = K.$$

The ordinary positive teaching dimension of L within $\mathcal{H}$ is also K.

Deterministic exact membership-query identification of $\mathcal{H}$ has worst-case query complexity K. One equivalence query suffices.

Relative to unrestricted MTSL_2°*,*

$$K \leq \mathrm{TD}(L) \leq K + f_{r,d}.$$

Proof. Only the points of W have variable labels. A hypothesis rejects either none or exactly one of them. A singleton is shattered, but two points cannot both be labeled negative. Thus VC dimension is one.

At any informative root query w, the negative branch contains only $L \setminus \{w\}$ and cannot split again. A depth-one tree exists, proving the Littlestone statement.

Every $w \in W$ must be taught positively to exclude its deletion. Those K examples suffice within $\mathcal{H}$.

On target L, a membership query distinguishes the competitor missing w only if the query is w. All K must therefore be queried, and querying them suffices. For equivalence queries, propose L. Either it is correct or the counterexample is the unique deleted word.

The full-class teaching lower bound follows from Lemma 2.4. For the upper bound, give all full paths as positives and all forbidden ordinary bigrams as negatives. Factoriality forces every consistent hypothesis to contain $\mathrm{Fac}(W) = L$, and the negative bigrams force it to be contained in L. □

There is also an exponential-dimensional subfamily, but it has a different description-size issue.

Proposition 8.2. *The class $\mathsf{MTSL}_2^\circ(\Sigma_{r,d})$ shatters $W_{r,d}$. Its VC and Littlestone dimensions are therefore at least d^r.*

Proof. For each $D \subseteq W_{r,d}$, conjoin the deletion constraint for every $w \in D$. The resulting language is $L_{r,d} \setminus D$. Thus every labeling of the K points is realized. Querying those points successively also shatters a binary tree of depth K. □

Some grammars in this Boolean cube have exponential size. In contrast, the dimension-one single-deletion family already gives the large characteristic and sampling lower bounds.

9 Sharp exact-identification bounds under uniform positives

For each $H \in \mathcal{H}$, sample independently and uniformly from the finite target language H. These distributions have full support. Let $t^*(r, d, \delta)$ be the smallest fixed sample size permitting exact

identification with error at most δ for every target in $\mathcal{H}$, allowing randomized learners and unlimited computation.

Put $V = |L_{r,d}|$. Counting layer intervals gives

$$V = 1 + \sum_{\ell=1}^{r} (r - \ell + 1) d^{\ell} + 2 \sum_{\ell=0}^{r} d^{\ell} + d^{r}. \tag{4}$$

In particular $V = \Theta(K)$, with absolute constants for $d \geq 2$: the internal-interval sum is at most $4K$, the two endpoint sums at most $4K$, and the remaining terms at most $2K$. For $d = 2$,

$$V = 9 \cdot 2^{r} - 2r - 5.$$

Theorem 9.1 (Minimax uniform-positive identification)**.** *Let*

$$a_V = -\log(1 - 1/V).$$

For $r \geq 1, d \geq 2$, and $0 < \delta \leq 1/4$,

$$t^*(r, d, \delta) \leq \left\lceil \frac{\log(K/\delta)}{a_V} \right\rceil,$$

and

$$t^*(r, d, \delta) \geq \frac{1}{a_V} \max \left\{ \log \frac{1}{2\delta}, \log\bigl(1 + 4K(1 - 2\delta)^2\bigr) \right\}.$$

Consequently,

$$t^*(r, d, \delta) = \Theta\left(d^{r} \left(r \log d + \log \frac{1}{\delta} \right) \right),$$

with absolute implicit constants.

Proof. For the upper bound, record the full paths observed. If exactly one full path is missing, output its deletion language; otherwise output L. The learner is correct whenever every full path belonging to the target has appeared.

Under the full target, each path has mass $1/V$. Under a deletion target, each remaining path has mass $1/(V - 1)$. The probability that some present full path is missed is at most

$$K(1 - 1/V)^{t}.$$

For the lower bound, let P_0 be uniform on L, and P_w uniform on $L \setminus \{w\}$. Under $P_0^{\otimes t}$, the probability of missing w is

$$q = (1 - 1/V)^{t}.$$

Conditional on that event, the observation law is exactly $P_w^{\otimes t}$. Thus the overlap of the two laws is q. Any test has error sum at least q, so simultaneous error at most δ requires $q \leq 2\delta$. This gives the first lower bound.

For the second, define the mixture

$$Q = \frac{1}{K} \sum_{w \in W} P_w^{\otimes t}.$$

Let Z count the full paths not observed in a sample from $P_0^{\otimes t}$. The likelihood ratio is

$$\frac{dQ}{dP_0^{\otimes t}} = \frac{Z}{Kq}.$$

Moreover,

$$\mathbb{E}_0 Z = Kq.$$

For distinct $u, w \in W$,

$$\mathbb{P}_0(u, w \text{ both unseen}) = (1 - 2/V)^t \le q^2.$$

The unseen indicators have nonpositive covariance, whence

$$\mathrm{Var}_0(Z) \le Kq(1-q).$$

Cauchy–Schwarz therefore gives

$$\mathrm{TV}(P_0^{\otimes t}, Q) = \frac{1}{2}\mathbb{E}_0 \left| \frac{Z}{Kq} - 1 \right| \le \frac{1}{2}\sqrt{\frac{q^{-1}-1}{K}}.$$

An identifier induces a test: decide $P_0^{\otimes t}$ exactly when its output is L. The two error probabilities are at most δ, including after averaging over the deletion targets. Hence

$$1 - 2\delta \le \mathrm{TV}(P_0^{\otimes t}, Q),$$

which implies

$$q^{-1} \ge 1 + 4K(1-2\delta)^2.$$

The argument applies to randomized learners by incorporating their independent random seed.

Finally,

$$V - 1 \le a_V^{-1} \le V,$$

and $V = \Theta(K)$. The maximum of the two logarithmic lower-bound terms is within absolute factors of $\log K + \log(1/\delta)$. □

For the unrestricted tier-feature learner, one can also compute the expected time exactly.

Proposition 9.2 (Exact coverage time)**.** *Under uniform sampling from $L_{r,d}$, the tier-feature learner identifies $L_{r,d}$ precisely when all K full paths have appeared. Its expected identification time is*

$$V H_K, \qquad H_K = \sum_{j=1}^{K} \frac{1}{j}.$$

With $\mu = K(1-1/V)^t$, its failure probability satisfies

$$\frac{\mu}{1+\mu} \le \mathbb{P}(\textit{failure after } t \textit{ draws}) \le \min\{1, \mu\}.$$

Proof. If a full path is absent, its deletion constraint remains consistent. If all appear, their factor closure is the entire target, so they witness every tier feature.

For the probability bound, $\mathbb{E}Z = \mu$ and $\mathbb{E}Z^2 \le \mu + \mu^2$. Cauchy–Schwarz yields $\mathbb{P}(Z > 0) \ge (\mathbb{E}Z)^2/\mathbb{E}Z^2$; the upper bound is the union bound. When j paths remain unseen, a new one appears with probability j/V. Summing the mean geometric waiting times V/j proves the expectation. □

10 Sparse sources, missing examples, noise, and approximation

10.1 Exact recovery needs an exposure parameter

Proposition 10.1 (No structure-only distribution-free rate)**.** *No finite sample bound depending only on alphabet size, locality, grammar size, DFA size, syntactic-monoid size, and confidence guarantees exact positive identification under arbitrary full-support sources, even for two fixed unary* SP_2 *languages.*

Proof. Take $L_0 = \{\varepsilon\}$ and $L_1 = \{\varepsilon, a\}$. Use point mass at ε for L_0, and masses $1-p, p$ on ε, a for L_1. Both sources have full support. Their t-sample laws have overlap $(1-p)^t$. Thus the sum of identification errors is at least this quantity. For fixed t, it approaches one as $p \downarrow 0$, while every structural parameter remains fixed. □

For a finite-feature target, define

$$p_f = \mathbb{P}_{X\sim P}\{f \in \phi(X)\}, \qquad f \in \phi(L).$$

The exact coverage principle gives

$$\mathbb{P}\{L_{\phi(S_t)} \neq L\} \leq \sum_{f\in\phi(L)} (1-p_f)^t. \tag{5}$$

If every $p_f \geq p_* > 0$, then

$$t \geq \frac{\log(|\phi(L)|/\delta)}{p_*}$$

suffices. Every live feature having positive probability is enough for almost-sure eventual identification, but not for a uniform rate.

Missing words need not be fatal if other words witness all required features. Missing a necessary distinction is fatal. In particular, a distribution supported on $L_{r,d} \setminus \{w\}$ is compatible with either that target or $L_{r,d}$ if full support is not required.

On arbitrary exhaustive positive texts, the necessary evidence may also be postponed for arbitrarily many repetitions. A characteristic-sample bound is not a bound on the convergence stage of every text.

10.2 Contamination

Suppose observations have distribution

$$\widetilde{P} = (1-\eta)P + \eta R,$$

where R is arbitrary.

Proposition 10.2 (An exponentially small contamination obstruction)**.** *For the uniform sources on* $L_{r,d}$ *and any single-word deletion, the contamination neighborhoods intersect whenever*

$$\eta \geq \frac{1}{V+1}.$$

At such a rate, exact identification with worst-case error below $1/2$ *is impossible, regardless of sample size.*

Proof. The two clean distributions have total variation distance $1/V$. For any discrete distributions P, Q,

$$\sum_x \max\{P(x), Q(x)\} = 1 + \mathrm{TV}(P, Q).$$

If $(1-\eta)(1+\mathrm{TV}(P,Q)) \le 1$, a probability distribution can be chosen to dominate both $(1-\eta)P$ and $(1-\eta)Q$: take their pointwise maximum and distribute the remaining mass arbitrarily. Subtracting either clean component and dividing by η supplies a valid contaminant. Thus the observed laws can coincide for different targets. □

A frequency gap restores a finite-feature guarantee.

Proposition 10.3 (Sufficient noise separation)**.** *Let the feature universe have size N. Suppose every live feature has clean probability at least p, and*

$$\Delta = (1-\eta)p - \eta > 0.$$

Threshold empirical feature frequencies at

$$\theta = \frac{(1-\eta)p + \eta}{2}.$$

Then

$$t \ge \frac{2}{\Delta^2} \log \frac{2N}{\delta}$$

suffices to recover the exact target with confidence $1-\delta$.

Proof. A forbidden feature has contaminated probability at most η; a live feature has probability at least $(1-\eta)p$. Each is separated from the threshold by at least $\Delta/2$.

For a Bernoulli variable Y, the second derivative of its centered log moment-generating function is a tilted Bernoulli variance, at most $1/4$. Integrating twice gives

$$\log \mathbb{E} e^{u(Y - \mathbb{E}Y)} \le u^2/8.$$

Exponential Markov inequality, optimized in u, yields

$$\mathbb{P}\{|\overline{Y} - \mathbb{E}Y| \ge z\} \le 2e^{-2tz^2}.$$

Apply this with $z = \Delta/2$, then union-bound over the features. Correct feature classification recovers $L_{\phi(L)} = L$. □

This is a sufficient finite-sample bound, not a claim of minimax optimality in the noise gap.

10.3 Approximation is different from exact identification

If error is measured only on a positive distribution and overacceptance is unrestricted, Σ^* has zero false-negative error without data. A meaningful alternative requires a conservative hypothesis $\widehat{L} \subseteq L$.

Proposition 10.4 (Standard one-sided approximation bound)**.** *For a finite feature universe of size N, clean positive data, and arbitrary P supported on L,*

$$L_{\phi(S_t)} \subseteq L,$$

and

$$\mathbb{P}\left\{P\big(L \setminus L_{\phi(S_t)}\big) > \epsilon\right\} \le 2^N e^{-t\epsilon}.$$

Proof. Observed features are contained in $\phi(L)$, proving containment. A fixed sublanguage omitting mass greater than ϵ contains all t examples with probability at most $e^{-t\epsilon}$. There are at most 2^N possible feature hypotheses. □

This is not ordinary two-sided labeled PAC learning. It also does not contradict Theorem 9.1, whose objective is exact equality.

11 Positive bounds under explicit restrictions

11.1 Bounded ambient tier width

Let $\mathcal{B}_b(\Sigma)$ permit arbitrary conjunctions of full-tier bigram constraints and constraints on tiers of size at most b. The restriction applies to *every hypothesis*, not just a chosen target description.

Proposition 11.1 (Explicit polynomial degree)**.** *For $0 \le b \le s$, an n-state target in $\mathcal{B}_b(\Sigma)$ has a characteristic sample satisfying*

$$|S| \le N_{s,b} := s^2 + \sum_{j=1}^{b} \binom{s}{j} j^2,$$

$$\ell(S) \le (3n-1)N_{s,b}.$$

For fixed b,

$$N_{s,b} \le s^2 + b^2(b+1)s^b.$$

The feature learner is computable in polynomial time in alphabet size and total input length for fixed b.

Proof. The displayed sum counts the available features. Apply Proposition 3.1 and the three-state occurrence monitor in Proposition 6.1. For the coarse bound, use $\binom{s}{j} \le s^j$, $j^2 \le b^2$, and $s^j \le s^b$. For fixed b, enumerate the permitted tiers and scan each projected example to compute all features. □

The canonical grammar of $L_{r,2}$ already has width at most four. Nevertheless its unrestricted characteristic cardinality is 2^r. There is no contradiction: its distinguishing competitors use wider tiers. A width promise about the target is weaker than a width restriction on the entire hypothesis space.

11.2 SL and SP benchmarks

The following results are consequences of the established feature framework, not claims of new qualitative learning theorems.

Proposition 11.2 (Fixed-locality baseline bounds)**.** *For SL_k°, characteristic cardinality is at most s^k, with total length at most ks^k.*

For SP_k, $k \ge 1$, characteristic cardinality is at most s^k, with total length at most ks^k, and

$$\mathrm{VCdim}(\mathsf{SP}_k) = s^k.$$

For the particular target $U_{k-1} = \Sigma^{\le k-1}$,

$$\chi_{\mathsf{SP}_k}(U_{k-1}) = s^{k-1}, \qquad \lambda_{\mathsf{SP}_k}(U_{k-1}) = (k-1)s^{k-1}.$$

Proof. For unmarked SL, every admitted k-factor is itself a positive word. Take one copy of each admitted factor.

For SP, the live features are $L \cap \Sigma^{\leq k}$, by subsequence downward closure. Choose its subsequence-maximal words. They cover every live feature and form a prefix-free set. Extending each to length k shows there are at most s^k; each has length at most k.

The set Σ^k is shattered by independently forbidding its members, proving the VC lower bound. For the upper bound, suppose $w_1, \ldots, w_m$ are shattered. For each i, a hypothesis rejecting only w_i on this set must forbid a subsequence u_i, of length at most k, occurring in w_i but in none of the other sample words. The u_i's form a subsequence antichain and hence a prefix-free set. Their sets of length-k extensions are nonempty and disjoint, so $m \leq s^k$.

The language U_{k-1} is obtained by forbidding every length-k subsequence. Each word u of length $k-1$ can be deleted alone by additionally forbidding u. Thus every such word is necessary. Together these words cover all shorter features and suffice. This includes $k = 1$, when the required sample is $\{\varepsilon\}$. □

The minimal complete DFA and syntactic monoid of U_{k-1} both have $k+1$ elements or states. They record length truncated at k, with multiplication

$$i \cdot j = \min\{i + j, k\}.$$

Different truncated lengths are distinguished by suitable suffix lengths. Its principal ideals are the distinct tails $\{i, \ldots, k\}$, so this monoid is commutative and $\mathcal{J}$-trivial. Thus varying locality already gives exponential positive characteristic requirements for SP with a linear-size monoid. Unlike the layered construction, its explicit forbidden grammar is itself exponentially large.

There is no inclusion ladder $\mathsf{SL} \to \mathsf{SP} \to \mathsf{TSL} \to \mathsf{MTSL}$. For instance, avoiding adjacent aa is local but not subsequence downward closed. Conversely, the language with at most one a over $\{a, b\}$ is SP_2, but not local of any fixed order: every fixed-length window of $ab^m a$, for sufficiently large m, occurs in a positive word.

12 Consequences for subregular transductions

12.1 An unknown-domain obstruction

For a regular language L, define the partial identity

$$\mathrm{id}_L(w) = w \quad (w \in L),$$

undefined elsewhere. A subsequential transducer realizes it by copying input letters and making exactly the domain-accepting states final.

Positive graph samples are finite sets of input–output pairs. Characteristic correctness means exact partial-function equality, including equality of domains, on every finite consistent extension.

Theorem 12.1 (Partial-identity transfer)**.** *Consider a function class containing*

$$\mathrm{id}_{L_{r,d}} \quad \textit{and} \quad \mathrm{id}_{L_{r,d}\setminus\{w\}} \qquad (w \in W_{r,d}).$$

Every positive graph characteristic sample for the central target contains all d^r pairs (w, w). Its input-plus-output length is at least

$$2(r+2)d^r.$$

The target has an $(r+4)$-state subsequential realization, and each competitor has a realization with at most $(r+4)^2$ states. Their domains use at most two nested tiers.

For the corresponding single-deletion family under uniform input sampling, the minimax bounds of Theorem 9.1 hold unchanged.

Proof. Deleting a domain word removes exactly one graph pair. The argument of Lemma 2.4 applies to graph inclusion.

Use the domain automata from Theorem 5.1 and Proposition 5.2, copying each letter on its transition and assigning empty final output precisely at accepting states. Lemma 4.1 supplies the domain descriptions.

Since the output is determined by the input and is the same in every hypothesis where defined, the statistical observation contains exactly the information in the positive input sample. □

This does not refute polynomial learning of total, fixed-locality ISL or OSL functions. It isolates the cost of learning an unknown domain in a larger hypothesis class.

12.2 A known synchronous controller

A complementary positive result makes the structural assumptions explicit.

Proposition 12.2 (Transition coverage and exact risk)**.** *Fix a known complete deterministic transition structure with n reachable states and input alphabet size s. Each transition has one unknown output symbol from an alphabet Γ, $|\Gamma| = g \geq 2$. There are no initial or final outputs.*

Let $M = ns$. A graph sample uniquely identifies the function if and only if its input paths traverse all M transition slots. A teaching sample exists with at most M pairs and total input length at most $nM = n^2 s$.

For independent inputs from a fixed distribution D, independent of the unknown table, let U_t count unvisited slots. The exact randomized minimax identification risk is

$$1 - \mathbb{E}[g^{-U_t}].$$

If every slot is visited by a random input with probability at least $p > 0$, then

$$t \geq \frac{\log(M/\delta)}{p}$$

suffices for error at most δ.

Proof. Aligned outputs reveal every visited transition label. If a slot is unvisited, changing only that label preserves the sample but changes the function on an input reaching that slot. Reachability therefore makes coverage necessary and sufficient.

A shortest path to a source state has length at most $n-1$. Append the slot's input letter to obtain a witness of length at most n. Choose one per slot.

Give the output labels independent uniform priors on Γ. Conditional on the sample, all unvisited labels remain independent and uniform. The best conditional success probability is g^{-U_t}. A learner guessing independent uniform labels on unvisited slots attains this probability for every target, so the Bayes lower bound is also the minimax value. Finally, a union bound gives $\mathbb{P}(U_t > 0) \leq Me^{-tp}$. □

For a known suffix-memory controller of order $k-1$, one may use

$$n = \sum_{j=0}^{k-1} s^j.$$

Unknown topology, output delays, and variable-length alignment are not covered by this proposition.

13 Remaining problems and scope of the conclusions

The strongest conclusion established here is a failure of uniform polynomial characteristic data, not a failure of bounded-class identification in the limit.

Problem 13.1 (Sharp alphabet exponent)**.** For $|\Sigma| = s$, determine

$$C(s) = \max_{L \in \mathsf{MTSL}_2^\circ(\Sigma)} \chi_{\mathsf{MTSL}_2^\circ(\Sigma)}(L).$$

The present results give, for $s \geq 5$,

$$3^{\lfloor (s-2)/3 \rfloor} \leq C(s) \leq s(s+1)2^{s-2},$$

with the stronger antichain upper bound in (3).

For the lower bound at arbitrary s, use three-symbol internal layers and add any leftover symbols as isolated singleton words, forbidding all bigrams involving them. The same individually deletable full paths remain. The optimal exponential rate is not determined here.

Problem 13.2 (Characteristic length under bounded ambient width)**.** For fixed tier width b, determine whether total characteristic length can be bounded polynomially in alphabet size and compact grammar size without including DFA size. Proposition 11.1 settles polynomial cardinality and gives a polynomial length bound when n is also a parameter.

The number of tiers alone is insufficient, and nesting alone is insufficient, by Theorem 4.2.

Problem 13.3 (Efficient construction of short certificates)**.** Determine structural restrictions under which the minimum positive-word cover of the families $\mathcal{A}_{xy}(L)$ can be computed efficiently. Distinguish output-size bounds from algorithms for constructing or optimizing the output.

For the single-bigram model, weighted transversal enumeration gives the exact certificate size. No general output-polynomial enumeration claim is established here.

Problem 13.4 (Proper single-tier and fixed-alphabet refinements)**.** Obtain sharp uniform characteristic cardinality and length bounds for unknown-tier TSL_k, refining the existing learners. Separately determine sharp bounds when the alphabet is fixed and locality varies. The main two-tier construction fixes locality and grows the alphabet.

Problem 13.5 (Sampling and genuine local transductions)**.** For explicit generative source families, bound distinguishing-feature probabilities in terms of source and grammar parameters. For total ISL, OSL, and restricted tier-based functions, obtain matching bounds that include topology, output delay, alignment, and source exposure. The unknown-domain reduction does not settle those problems.

A necessary-and-sufficient criterion based only on the abstract target monoid is not proved and would not explain the class-relative separation exhibited here. A successful enriched criterion would have to account for the permitted refinements, the alphabet morphism and accepting structure, and, for random data, the probabilities of their distinguishing events.

Conclusion. Two nested tiers already permit exponentially many individually necessary positive witnesses while the target has linearly many states and a quadratic-size $\mathcal{J}$-trivial syntactic monoid. A complementary sparse grammar realizes the weighted number of minimal hypergraph transversals exactly. These are explicit, rigorous obstructions beyond the qualitative statement that a class is learnable. Their statistical counterparts show why full support is not a quantitative exposure guarantee, and why low VC or Littlestone dimension does not ensure inexpensive exact recovery from positive evidence.

References

[1] D. Angluin. Inductive inference of formal languages from positive data. *Information and Control*, 45(2):117–135, 1980.

[2] C. de la Higuera. Characteristic sets for polynomial grammatical inference. *Machine Learning*, 27:125–138, 1997. doi:10.1023/A:1007353007695.

[3] J. Heinz. String extension learning. In *Proceedings of the 48th Annual Meeting of the Association for Computational Linguistics*, pages 897–906, 2010.

[4] J. Heinz, A. Kasprzik, and T. Kötzing. Learning in the limit with lattice-structured hypothesis spaces. *Theoretical Computer Science*, 457:111–127, 2012.

[5] A. Jardine and J. Heinz. Learning tier-based strictly 2-local languages. *Transactions of the Association for Computational Linguistics*, 4:87–98, 2016.

[6] A. Jardine and K. McMullin. Efficient learning of tier-based strictly k-local languages. In *Language and Automata Theory and Applications*, Lecture Notes in Computer Science 10168, pages 64–76. Springer, 2017.

[7] L. Swanson. Learning multi tier-based strictly 2-local languages. In *Proceedings of the 18th Meeting on the Mathematics of Language*, pages 33–43, 2025.

[8] L. Swanson. Syntactic learning over tree tiers. *Logics*, 4(2), article 5, 2026. doi:10.3390/logics4020005.

[9] J. Fu, J. Heinz, and H. G. Tanner. An algebraic characterization of strictly piecewise languages. In *Theory and Applications of Models of Computation*, Lecture Notes in Computer Science 6648, pages 252–263. Springer, 2011.

[10] D. Lambert. On the structure of sets testable in the strict sense. In *Proceedings of the 18th Meeting on the Mathematics of Language*, pages 59–73, 2025.

[11] T. Eiter and G. Gottlob. Identifying the minimal transversals of a hypergraph and related problems. *SIAM Journal on Computing*, 24(6):1278–1304, 1995.

[12] J. Chandlee, R. Eyraud, and J. Heinz. Learning strictly local subsequential functions. *Transactions of the Association for Computational Linguistics*, 2:491–504, 2014.

[13] J. Chandlee, R. Eyraud, and J. Heinz. Output strictly local functions. In *Proceedings of the 14th Meeting on the Mathematics of Language*, pages 112–125, 2015.

[14] P. Burness and K. McMullin. Learning multiple independent tier-based processes. In *Proceedings of the 15th International Conference on Grammatical Inference*, Proceedings of Machine Learning Research 153, pages 66–80, 2021.

[15] K. Wu and J. Heinz. String extension learning despite noisy intrusions. In *Proceedings of the 16th International Conference on Grammatical Inference*, Proceedings of Machine Learning Research 217, pages 80–95, 2023.